# International Capital Flows in the Context of Financial Stability

## 1MT404 Financial Stability

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# Outline


1. Macroeconomic Accounting & Sectoral Balances
2. The Intertemporal Approach to the Current Account
3. Applications and Limitations
4. Critiques of the Intertemporal Approach: Gross vs. Net Flows
5. Global Patterns
6. The Eurozone Crisis: A Flawed Optimum Currency Area
7. Managing Capital Flows: The Policy Toolkit
8. Geopolitics, Public Debt, and De-Dollarization
9. Conclusions

# Balance of Payments (BoP) Framework (BPM6)

## The fundamental identity

The BoP records every transaction between residents and non-residents twice (double entry):

$$CA + KA + NEO = FA$$

$FA =$ net acquisition of foreign assets minus net incurrence of foreign liabilities (net lending to the rest of the world); $NEO =$ net errors and omissions.

**A current account deficit** is not "lost trade": it is the counterpart of a net capital inflow – a net increase in liabilities to non-residents.

- **Current account (CA)**: goods, services, primary income (investment returns), secondary income
- **Capital account (KA)**: capital transfers, non-produced non-financial assets – small, except EU investment grants in Central Europe
- **Financial account (FA)**: FDI, portfolio investment, other investment (loans, deposits), reserve assets

$FA$ is a flow; the *stock* of net foreign assets also moves with prices and exchange rates (valuation effects, NIIP slides).

# Sectoral Balances and the Macroeconomic Identity

From the national income identity $Y = C + I + G + (X - M)$ we derive the sectoral financial balances:

$$(S_p - I_p) + (T - G) = (X - M) \approx CA$$

- $(S_p - I_p)$: **private sector balance**
- $(T - G)$: **public sector balance** (government surplus or deficit)
- $(X - M)$: **foreign sector balance**; the current account adds net primary and secondary income

**Core implication:** a current account deficit ($CA < 0$) *must* be matched by a private sector deficit (e.g. US household borrowing before 2008), a public sector deficit, or both. It is a saving–investment gap, not just a trade phenomenon.

# The Kalecki Profit Equation: An Accounting Lens

Assuming workers consume what they earn ($C_w = W$), aggregate profits ($P$) are

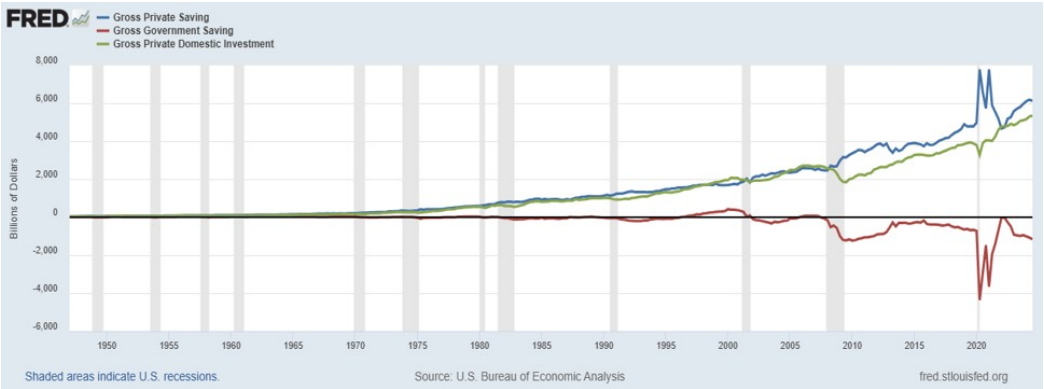
$$P = I + (G - T) + (X - M) + C_k - S_w$$

Profits = investment + government deficit + current account surplus + capitalist consumption − workers' saving.

- **Export-led economies (Germany, China):** the $(X - M)$ term supports profits without domestic public or private deficits
- **Deficit economies (US, UK):** with $(X - M) < 0$, profits are supported by government deficits $(G - T)$ or by debt-financed investment and consumption

An identity, not a causal model: it shows where profits come from in the accounts, not what drives what (Kalecki; Lavoie 2014).

# US Gross Private Saving, Investment and Government Saving




**Takeaway:** Since 2008 US private saving has exceeded private investment, and the gap mirrors the government deficit – the sectoral identity at work.

*Source: FRED (gross private saving, gross government saving, gross private domestic investment; USD billion, quarterly).*

# The Intertemporal Approach: Core Theory

The intertemporal approach (IA) views the current account as the optimal, forward-looking outcome of consumption-smoothing and investment decisions (Obstfeld and Rogoff 1996).

**Intertemporal budget constraint (IBC):** with $B_t$ = net foreign *assets* at the end of period $t$ and $TB$ = trade balance,

$$(1+r)B_{t-1} + \sum_{s=t}^{\infty} \frac{TB_s}{(1+r)^{s-t}} = 0$$

A debtor ($B_{t-1} < 0$) must run future trade surpluses whose present value repays its net foreign debt, $-(1+r)B_{t-1}$ – no Ponzi schemes.

**Consumption smoothing:**

- **Temporary shocks:** a temporary fall in income (e.g. a natural disaster) leads to a CA deficit as the country borrows to smooth consumption
- **Permanent shocks:** a permanent fall in income leads to an immediate cut in consumption, leaving the CA largely unchanged

# Two-Period Small Open Economy Model

Two periods, one good, exogenous world rate $r$, endowments $Y_1, Y_2$, $U = u(C_1) + \beta u(C_2)$; $B_1 =$ net foreign assets after period 1, with $B_0 = B_2 = 0$.

Budget constraints:

$$C_1 + B_1 = Y_1, \qquad C_2 = Y_2 + (1+r)B_1$$

Intertemporal budget constraint:

$$C_1 + \frac{C_2}{1+r} = Y_1 + \frac{Y_2}{1+r}$$

Utility maximisation $\Rightarrow$ Euler equation:

$$u'(C_1) = \beta(1+r)\,u'(C_2)$$

If $\beta(1+r) = 1$, consumption is flat (any concave $u$):

$$C_1 = C_2 = \frac{(1+r)Y_1 + Y_2}{2+r}$$

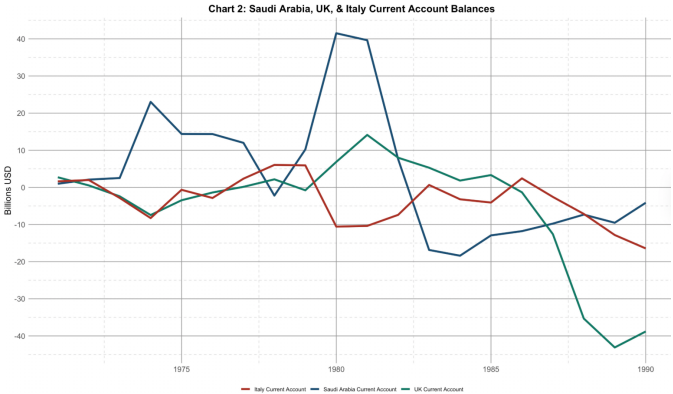
$$CA_1 = Y_1 - C_1 = B_1$$

A surplus when $Y_1$ is temporarily high; a deficit when $Y_1$ is temporarily low.

# Applications: Oil Shocks, Immigration and Debt Crises

- **1970s oil shocks:** exporters (Saudi Arabia) ran large surpluses and lent them abroad; importers borrowed to smooth consumption (next slide)
- **Israel, 1990s immigration wave:** a current account deficit driven by *investment* – housing and capital for about a million immigrants – the IA's investment channel
- **US–China imbalances:** the US deficit reflects low saving. China's surplus is a *puzzle* for the IA: a fast-growing economy should borrow against its future income, not lend – the "allocation puzzle" (Gourinchas and Jeanne 2013)
- **Sovereign debt:** debt is sustainable only if the future surpluses it requires are feasible; Greece needed surpluses but ran deficits

# Intertemporal Approach Example: The Oil Shocks of the 1970s




**Takeaway:** Saudi Arabia saved its windfalls (surpluses of about \$23 bn in 1974 and \$41 bn in 1980) and borrowed once oil prices fell; Italy borrowed after the shocks – smoothing in practice.

*Source: Benigno, G. (2025), "Background note on trade and trade imbalances", Substack.*

# Current Account Surpluses in Developed Economies

Structural ("natural") drivers:

- Ageing populations $\Rightarrow$ high private saving
- A positive net international investment position ("rentier" status) $\Rightarrow$ net investment income
- A high capital/labour ratio $\Rightarrow$ diminishing returns to domestic investment
- Technological leadership and high value-added exports
- Low investment relative to saving

These create persistent surpluses without policy intervention – although policy (wage restraint, fiscal rules, exchange rate regimes) can reinforce them.

# The Lucas Paradox and "Uphill" Capital Flows

**The paradox (Lucas 1990):** capital should flow to capital-scarce poor countries, where returns should be higher – but far too little does. **Strong form:** net flows go "uphill", from poorer to richer countries (Prasad, Rajan and Subramanian 2007).

**Why?**

- Lower risk-adjusted returns: weaker institutions, human capital, sovereign risk (Alfaro, Kalemli-Ozcan and Volosovych 2008)
- Official flows: reserve accumulation by EM central banks drives the uphill flow; private capital still flows downhill (Alfaro et al. 2014)
- Faster-growing EMs receive *less* capital (Gourinchas and Jeanne 2013)

**On the receiving side – why rich deficit countries borrow:**

- Falling household saving; consumption growth above productivity growth
- Low real rates and rising household and public debt
- Booms in non-productive investment (housing, real estate)
- Rising unit labour costs and lost competitiveness

# From Theory to Crisis: What the IA Misses

**So far:** the IA explains why countries run surpluses or deficits as rational, forward-looking behaviour.

**The problem:** it models *net* resource transfers. Persistent current account deficits signal a **medium- to long-term vulnerability** (structural fragility), but they are rarely the immediate cause of a crisis.

Modern financial crises are almost always triggered by **sudden stops in gross financial flows** – bank lending, wholesale funding, portfolio flows – dynamics that the current account balance does not capture.

# Two Views of Global Imbalances Before 2008

| | **Global saving glut** (Bernanke 2005) | **Global banking glut / gross flows** (Bernanke et al. 2011; Shin 2012; Borio and Disyatat 2011) |
|---|---|---|
| Story | Excess saving in Asia and oil exporters flows to the US, lowers long-term rates and fuels the housing boom | European banks borrow dollars in the US and buy US mortgage securities; gross flows, not net, finance the boom |
| Key variable | Current account (net flows) | Gross positions, funding and leverage |
| Evidence | Large Asian surpluses and reserve accumulation; low long-term yields (the "conundrum") | The euro area's current account was close to balance, yet European investors bought a large share of US private-label mortgage securities |
| Policy | Rebalance saving and demand across countries | Monitor and regulate gross exposures, dollar funding and bank leverage |

**Takeaway:** Both views capture part of the story – but only the gross-flow view explains why European banks, not Asian central banks, needed dollar support in 2008.

# The Breakdown of the Intertemporal Approach

While the IA models **real resource transfers** (net flows) well, it misses the mechanics of modern financial crises.

**Critique by Borio, Disyatat and Shin:**

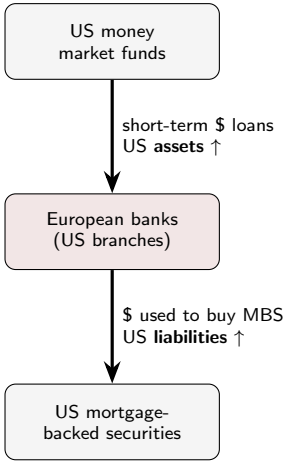
- **Saving is not financing:** the IA conflates saving (an ex-post national accounting residual – income not consumed) with financing (an ex-ante cash-flow concept – access to purchasing power)
- **Endogenous money:** banks do not merely intermediate pre-existing saving (the loanable funds view); they create purchasing power when they lend. Investment can be financed before saving exists, which decouples credit booms from the current account
- **The "excess elasticity" of credit:** fragility stems from the financial system's ability to expand balance sheets without a firm anchor in saving – not from a global saving glut

# Gross vs. Net Capital Flows: The 2008 Example

Before 2008 the euro area's current account was close to balance, but **gross flows** with the US exploded:

- US money market funds lent dollars to European banks $\Rightarrow$ **US foreign assets** rise (claims on European banks)
- European banks used the dollars to buy US mortgage-backed securities $\Rightarrow$ **US foreign liabilities** rise
- Net flow $\approx 0$; gross positions large and maturity-mismatched (short-term funding, long-term assets)

When mortgage values fell in 2007–08, the money funds pulled back: a **sudden stop** in dollar funding for European banks, contained only by Fed swap lines (FX lecture, slide 81).




net flow $\approx 0$, gross flows large

# Gross Flows: Surges, Stops, Flight and Retrenchment

| | | |
|---|---|---|
| Gross **inflows** (by foreigners) | **Surge** sharp rise | **Stop** sharp fall |
| Gross **outflows** (by residents) | **Flight** sharp rise | **Retrenchment** sharp fall |

Forbes and Warnock (2012)

- Episodes are driven mainly by **global factors** – risk aversion, global growth – and spread through contagion; domestic factors matter less
- **Sudden stop** (Calvo 1998): an abrupt fall in net inflows forces the current account to adjust – a crisis when debt is short-term and in foreign currency

## What a central bank monitors

Gross inflows and outflows by type and investor; short-term external debt relative to reserves; currency mismatches; non-resident holdings of local-currency bonds; rollover of bank funding.

# What the Data Tell Us: A Road Map

Four empirical regularities to keep in mind

## What we are looking for

Theory predicts that capital flows to where returns are highest. The data show a more complex picture.

1. **FDI is the anchor.** It is the least volatile component of capital flows and the least linked to sudden stops.
2. **Global imbalances are persistent.** The US has run a current account deficit almost every year since the early 1980s; East Asia has run surpluses.
3. **Saving rates explain the surplus side.** East Asia & Pacific saves about 35% of GDP (China over 40%); North America under 20%.
4. **Valuation effects matter as much as flows.** The US NIIP moves with the dollar and equity prices: between end-2024 and mid-2026 deficits of about \$1.6 trillion moved it by only \$0.3 trillion.

# Empirical Patterns: FDI Inflows (1990–2023)

## By development level

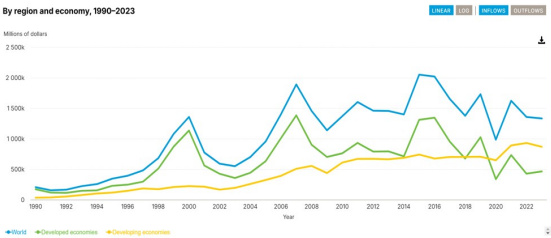
FDI went mostly to developed economies until the mid-2000s; flows to developing economies (above all Asia) have grown steadily since.

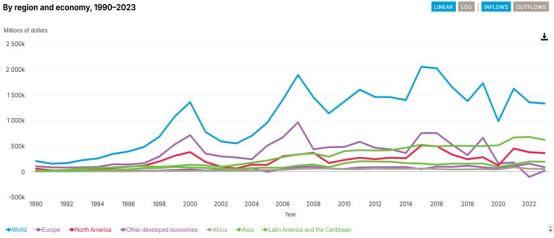
## By region

- Asia: strong growth (China, then India and South-East Asia)
- Europe and North America: the largest but most volatile recipients
- Africa, Latin America and the Caribbean: smaller, commodity-driven

FDI is the *least volatile* component of capital flows. Portfolio and "other" flows (bank and wholesale funding) dominate volatility and crisis transmission.

# FDI Inflows, 1990–2023

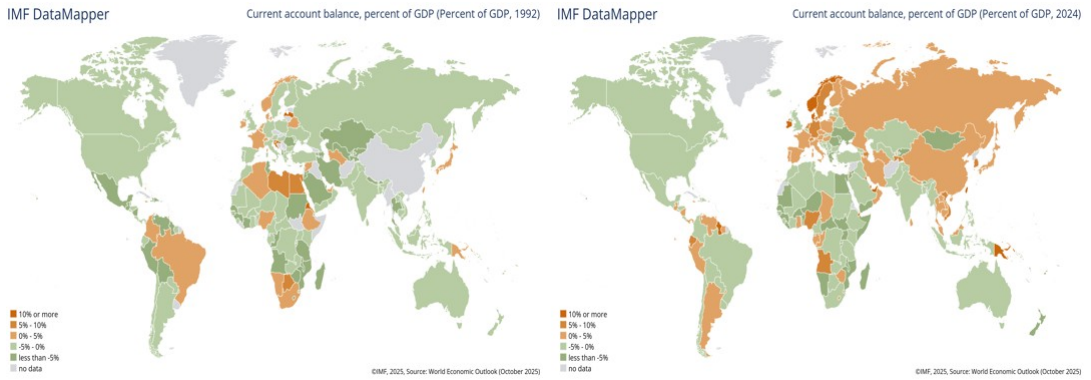



**Takeaway:** World FDI peaked in 2007 and 2015–16 and has been weaker since; flows to developing economies (left, yellow) have been the steadiest.

*Source: UNCTAD, World Investment Report 2024 / UNCTADstat; FDI inflows, USD million; by development level (left) and by region (right).*

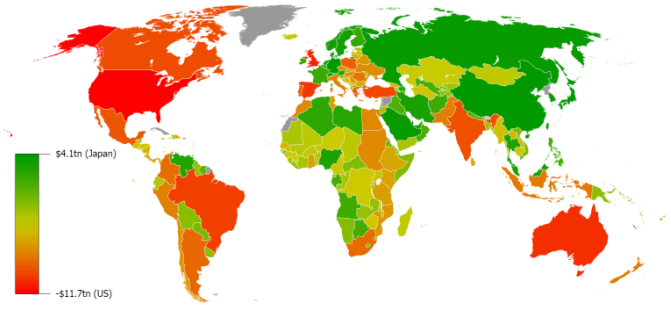
# Current Account Balances, 1992 and 2024




**Takeaway:** By 2024 surpluses (orange) are concentrated in China, northern Europe and commodity exporters, while the US and most of Africa and Latin America run deficits (green).

*Source: IMF, World Economic Outlook (October 2025), via IMF DataMapper; current account balance, % of GDP.*

# Cumulative Current Account Balances




**Takeaway:** Over the period shown, the US cumulative deficit reached about \$11.7 trillion and Japan's cumulative surplus about \$4.1 trillion – the two ends of the colour scale.

*Source: D. Wales, lecture notes.*

# The Persistent US Current Account Deficit

- A deficit almost every year since the early 1980s, widest in 2006 (about 6% of GDP); in Q2 2026 $246 bn, or 3.0% of GDP (BEA)
- Financed by foreign purchases of US assets: as issuer of the main reserve currency, the US borrows in its own currency and needs no reserves
- Counterparts: low household and public saving, booms in non-tradables, asset price booms partly fuelled by inflows, a shrinking manufacturing export base

Deficits reflect saving–investment gaps, not "unfair trade" – although trade policy can shift where the deficit shows up bilaterally (next slides).

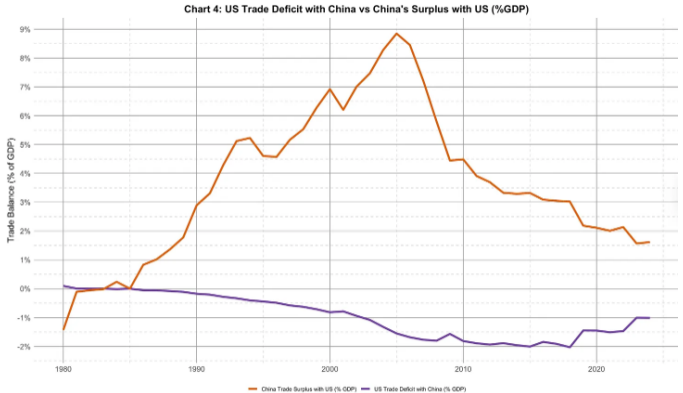
# US and China: Current Account and Trade Balances




**Takeaway:** China's current account surplus peaked near 10% of GDP in 2007 and has been 1–3% since; the US deficit peaked near 6% in 2006.

*Source: Benigno, G. (2025), "Background note on trade and trade imbalances", Substack.*

# US–China Bilateral Trade Balance




**Takeaway:** Measured against China's GDP, its surplus with the US fell from about 9% (2006) to under 2% (2024); against US GDP the bilateral deficit is now about 1%.

*Source: Benigno, G. (2025), "Background note on trade and trade imbalances", Substack.*

# Asian Economies Financing Global Deficits

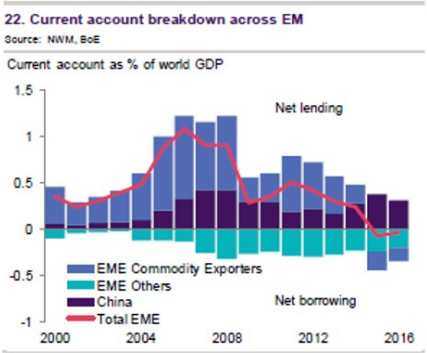
China and Asian emerging markets:

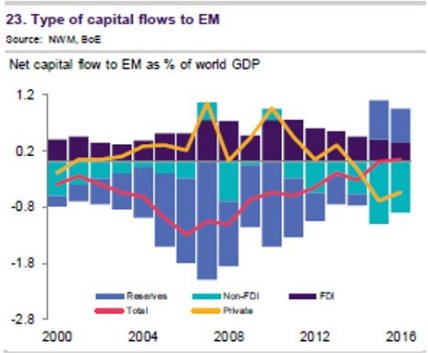
- High private saving and export-led growth (productivity surges)
- Massive reserve accumulation (self-insurance after 1997, and competitiveness through managed exchange rates)
- *Public* (not private) financing of developed-world deficits
- Young demographics initially boosted saving

Consequences:

- US debt was financed largely by EM official reserves – the "uphill" flow of the Lucas-paradox slide
- Global asset ownership shifted toward EM central banks and sovereign wealth funds

# Emerging Markets: Current Accounts and Types of Capital Flows

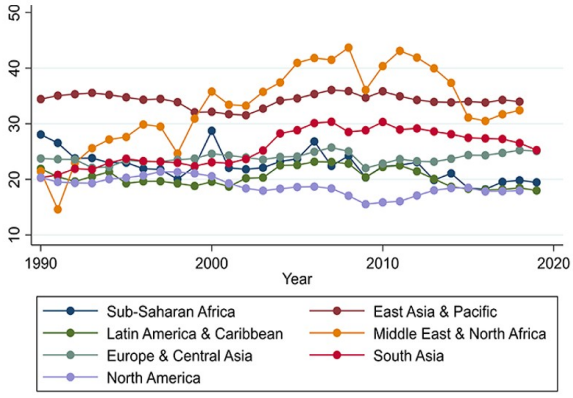



**Takeaway:** Before 2008 EM surpluses (mostly China's) were recycled as reserve accumulation – official outflows (blue, right) – while FDI and private inflows continued.

*Source: NatWest Markets; Bank of England.*

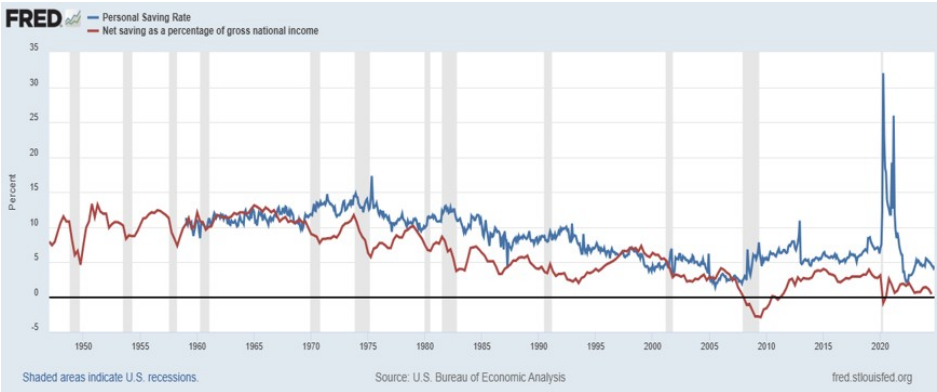
# Saving Rates by Region




**Takeaway:** East Asia & Pacific saves about 35% of GDP and South Asia 25–30%, against under 20% in North America; MENA swings with oil prices.

*Source: World Bank, World Development Indicators, gross savings (% of GDP), regional aggregates.*

# US Saving Rates

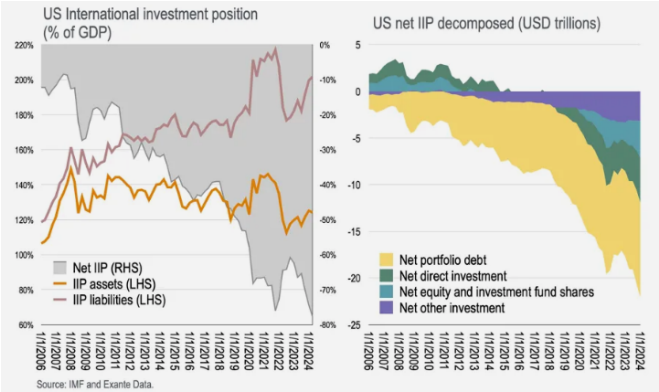



**Takeaway:** US net national saving fell from about 10% of GNI in the 1960s to around zero by 2008 and has stayed low; the 2020 spike in household saving was temporary.

*Source: FRED (personal saving rate; net saving as % of gross national income).*

# US International Investment Position

## Exorbitant privilege?

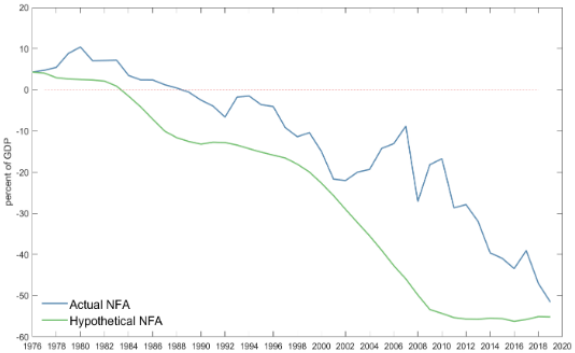



**Takeaway:** The US net position fell from about $-10\%$ of GDP (2007) to about $-80\%$ (2024), mostly because foreign holdings of US debt and equity securities grew.

*Source: Exante Data (J. Nordvig), Money Inside Out (Substack), "United States: changing net IIP"; data: IMF.*

# The US NIIP and Valuation Changes

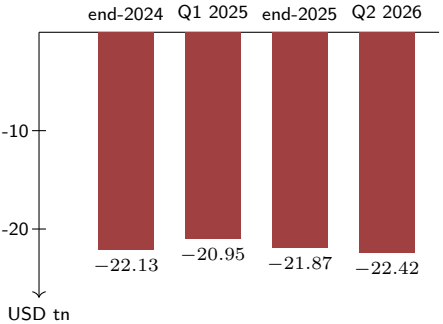
## Actual vs. hypothetical NIIP without valuation changes since 1976




**Takeaway:** Until about 2010 valuation gains kept the US position far less negative than cumulated deficits imply; by 2019, when this chart ends, the gap had closed.

*Source: G. Hall, Brandeis University, Econ 303 notes, "The NIIP paradox"; data: BEA.*

# 2024–2026: Valuation at Work Again




US net international investment position, end of period (revised data)

- Current account deficits: $1.12 trillion in 2025 and about $0.46 trillion in the first half of 2026 – about $1.6 trillion in total
- Yet the NIIP fell by only about $0.3 trillion
- The difference, about $1.3 trillion, is mostly valuation: the weaker dollar in 2025 raised the dollar value of US-owned foreign assets

**Takeaway:** The NIIP is not cumulated deficits: over 18 months valuation offset about four-fifths of the deficits.

*Source: U.S. Bureau of Economic Analysis, international transactions and IIP releases (June and September 2026); series IIPUSNETIQ via FRED.*

# NIIP: Beyond Trade Balances

The net international investment position (NIIP) is the value of US-owned foreign assets minus foreign-owned US assets.

$$\Delta NIIP_t = CA_t + \Delta VA_t + RES_t$$

where $CA$ is the current account, $\Delta VA$ the **valuation changes** and $RES$ other changes in volume.

- **Price effects:** if US stock markets outperform foreign markets, US *liabilities* rise faster than assets and the NIIP falls
- **Exchange rate effects ("exorbitant privilege"):** US liabilities are mostly in dollars, US foreign assets largely in foreign currencies – a **weaker dollar** raises the dollar value of the assets and mechanically improves the NIIP (Gourinchas and Rey 2007)

**Takeaway:** In 2020–24 US equity outperformance made the NIIP more negative; in 2025 the weaker dollar reversed part of it.

# From Global Patterns to a Case Study

**Pattern established:** gross flows dominate volatility; FDI is stable, portfolio and banking flows are not.

**Next:** the euro area (1999–2012) is the clearest case where a sudden stop in gross banking flows – acting on the medium-term vulnerability of divergent unit labour costs – triggered a systemic crisis.

# The Eurozone Debt Crisis: Gross Flows and the Real Exchange Rate

The euro area is a textbook case of capital flow misallocation within an incomplete optimum currency area (OCA).

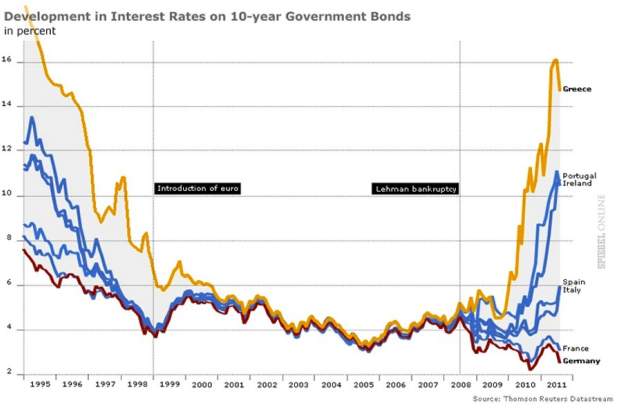
**The setup (1999–2007):**

- Monetary union removed nominal exchange rate risk and compressed sovereign spreads:

$$i^{\text{Periphery}} \downarrow \rightarrow \; i^{\text{Periphery}} \approx i^{\text{Core}}$$

- Large **gross capital flows** from the core (e.g. Germany, France) to the periphery (e.g. Spain, Greece), intermediated by banks
- The inflows financed domestic demand booms – Spain: housing and construction; Greece: public consumption and fiscal expansion
- Weak allocation $\rightarrow$ resources moved into non-tradables

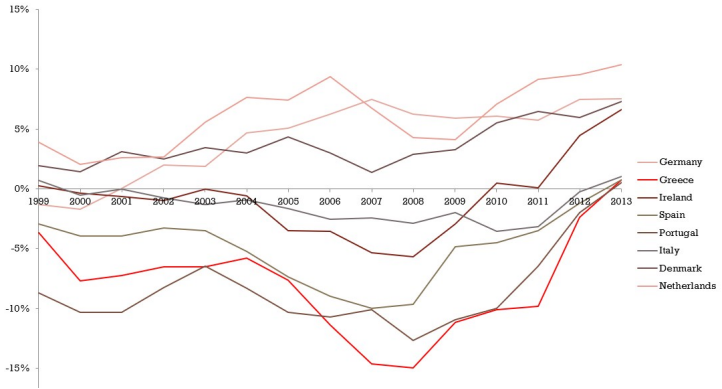
# Ten-Year Government Bond Yields, 1995–2011




**Takeaway:** Spreads vanished after the euro's introduction – markets priced Greek and German debt almost alike – and reopened only after 2008.

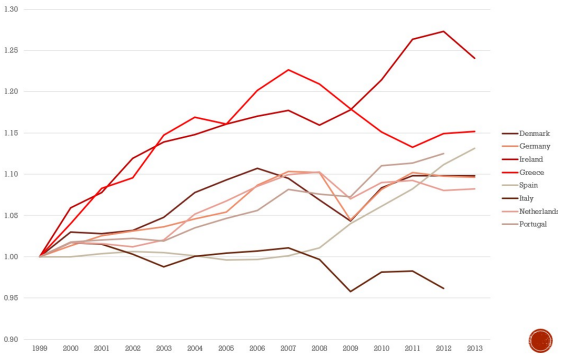
*Source: Spiegel Online; data: Thomson Reuters Datastream.*

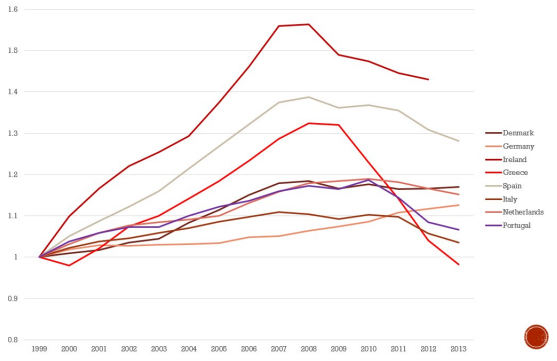
# Current Account Balances, 1999–2013




**Takeaway:** Greece, Portugal and Spain ran deficits of 10–15% of GDP by 2008 while Germany and the Netherlands moved into large surpluses; the deficits closed only through the 2010–13 contraction.

# Productivity and Consumption, 1999 = 1




**Takeaway:** Demand ran ahead of productivity in Ireland, Spain and Greece: consumption (right) rose 30–55% between 1999 and 2008, then fell back after 2010.

# Unit Labour Costs (ULC) and Cost Pressures

Nominal unit labour cost ($w$ = nominal wage, $L$ = labour, $Y$ = real output):

$$ULC_n = \frac{wL}{Y} = \frac{w}{Y/L}$$

- $ULC_n$ rises when nominal wage growth ($\Delta w$) exceeds productivity growth ($\Delta a$):

$$\Delta ulc_n \approx \Delta w - \Delta a$$

- **Periphery:** large capital inflows pushed up wages without matching productivity gains: $\Delta w \gg \Delta a \Rightarrow ULC_n \uparrow$
- **Core (e.g. Germany):** wage moderation kept costs in check: $\Delta w \approx \Delta a \Rightarrow ULC_n$ stable

# Real Unit Labour Costs and the Labour Share

Real ULC ($ULC_r$) deflates nominal ULC by the price level ($P$):

$$ULC_r = \frac{ULC_n}{P} = \frac{wL}{PY} = \frac{w/P}{Y/L}$$

- Real ULC is the ratio of the **real wage** ($w/P$) to **labour productivity** ($Y/L$)
- Economically, it is the **labour share of income** – the part of national income paid to workers
- In growth rates: $\Delta ulc_r \approx \Delta w_{real} - \Delta a$

**For the euro crisis:** nominal ULC measured the loss of export competitiveness; real ULC tracked whether workers or capital owners captured the gains of the boom.

# The Real Exchange Rate Channel

With fixed nominal exchange rates inside the euro area, cost competitiveness depends on relative nominal ULCs:

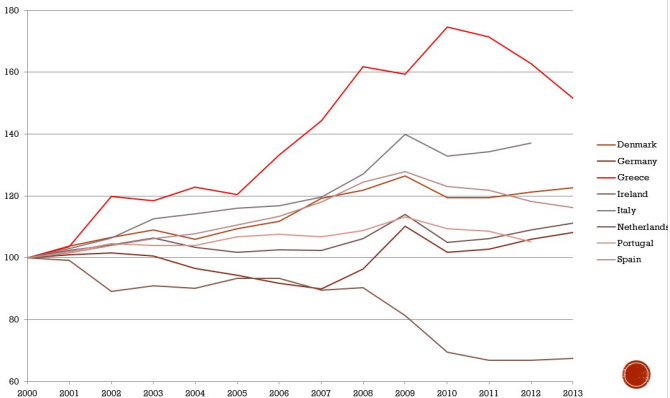
$$REER \sim \frac{ULC^{\text{Periphery}}}{ULC^{\text{Core}}}$$

- Rising relative ULC $\rightarrow$ **real appreciation** $\rightarrow$ lost competitiveness (exports $\downarrow$, imports $\uparrow$) $\rightarrow$ current account deficits financed by gross inflows

**Crisis dynamics (2010 onwards):**

- A sudden stop in gross private flows $\rightarrow$ funding crisis in the banking systems
- No nominal devaluation possible (monetary union)
- Adjustment through **internal devaluation**: wage compression, high unemployment and severe contraction

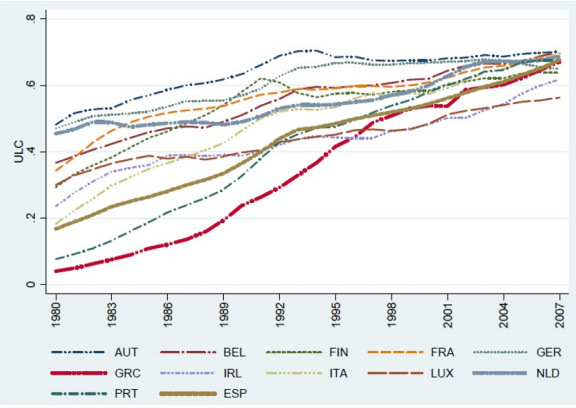
# ULC Divergence: Core vs. Periphery (Nominal)




**Read-off:**

- Greece's nominal ULC: about $+60\%$ by 2008 and $+75\%$ by 2010 ($2000 = 100$); Germany flat or falling
- Mechanism: capital inflows $\Rightarrow$ wage growth $\gg$ productivity growth
- No devaluation possible $\Rightarrow$ competitiveness loss baked in

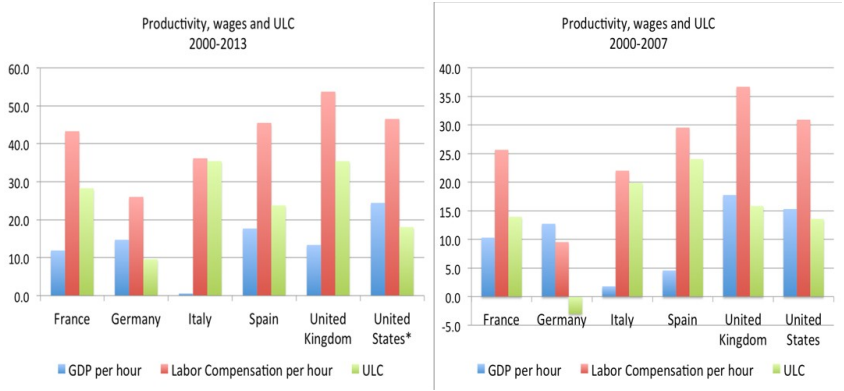
# Unit Labour Costs over the Longer Run, 1980–2007




**Takeaway:** Over 1980–2007 ULCs rose fastest in Greece, Portugal and Spain – from low starting levels – converging on the core by the late 2000s.

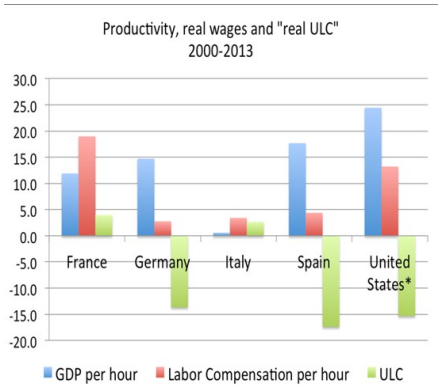
*Source: Felipe, J. and Kumar, U. (2011), "Unit labor costs in the Eurozone: the competitiveness debate again", Levy Economics Institute WP 651; data: OECD.*

# Productivity, Wages and ULC, 2000–2013 and 2000–2007




**Takeaway:** In 2000–07 German compensation per hour rose less than productivity, so its ULC fell; in Spain and Italy compensation outran productivity by about 20 percentage points.

# Productivity, Real Wages and Real ULC, 2000–2013




**Takeaway:** Deflated by prices, real ULC fell by about 14% in Germany and 17% in Spain: the labour share fell even where nominal ULC rose.

# The "Competitiveness Paradox" (2000–2007)

In the periphery's boom years, **nominal ULC surged while real ULC stayed flat or fell.**

**The mechanism:**

- Capital inflows drove domestic demand $\rightarrow$ high non-tradable inflation ($P \uparrow$)
- Tight labour markets pushed up nominal wages ($\Delta w > 0$)
- With stagnant productivity ($\Delta a \approx 0$), **nominal ULC surged**

**Why real ULC fell ($ULC_r = ULC_n / P$):**

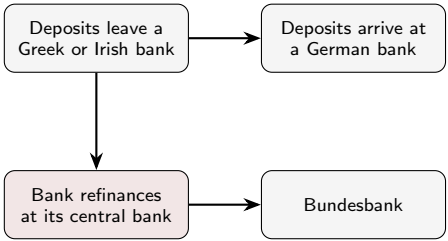
- Firms had pricing power and raised prices faster than wages, so $P$ grew faster than $ULC_n$

**The policy error:**

- Real ULC (the labour share) suggested labour was getting "cheaper" relative to output
- But rising nominal ULC was pricing these countries out of export markets and building external imbalances

# TARGET2: When Private Money Left, the Eurosystem Stepped In

- When private lenders and depositors pulled out of periphery banks, the banks refinanced at their national central bank
- The payments ran through TARGET2: the periphery central bank's **liability** to the Eurosystem rose; the Bundesbank's **claim** rose
- April 2012: periphery central banks' TARGET2 liabilities **€851 bn**; Bundesbank claims **€644 bn** (Lubik and Rhodes 2012)
- In Ireland the rise in TARGET2 liabilities was a multiple of the current account deficit – capital flight, not trade (Buiter et al. 2011)




TARGET2: periphery central bank **liability** ↑, Bundesbank **claim** ↑

**Takeaway:** Inside a currency union a sudden stop shows up as official (Eurosystem) financing replacing private flows – a funding crisis, not a currency crisis; TARGET2 balances are a symptom, not the cause.

# The Eurozone Policy Response (2010–2015)

## What was done

**Crisis management (short-run firefighting):**

- **EFSF (2010) / ESM (2012):** conditional loans to Greece, Ireland, Portugal and Cyprus, and to Spain for bank recapitalisation; conditionality tied to fiscal consolidation and structural reform
- **ECB Securities Markets Programme (SMP, 2010):** sterilised secondary-market purchases of periphery sovereign bonds
- **ECB Outright Monetary Transactions (OMT, 2012):** Draghi's "whatever it takes" backstop – *unlimited*, conditional sovereign bond purchases

**Structural reforms (medium run):**

- **Banking union:** Single Supervisory Mechanism at the ECB (2014) and Single Resolution Mechanism (2016) – weakened the *bank–sovereign doom loop* by moving supervision and resolution above the national level
- **Fiscal Compact (2012):** treaty-level structural deficit limits with automatic correction mechanisms

# The Eurozone Policy Response – What Remains Unresolved

## Incomplete architecture

The euro area fixed the acute crisis but left root structural flaws largely in place.

- **No European Deposit Insurance Scheme (EDIS):** national deposit guarantee schemes remain, preserving the bank–sovereign link at the margin; the third pillar of banking union is still not agreed
- **No permanent fiscal union or common safe asset:** NextGenerationEU (2020) showed that large joint borrowing is possible, but it is temporary; there is no permanent euro-area safe asset that recycles surplus savings automatically (as US Treasuries do)

**Takeaway:** The response *stabilised* the crisis but did *not* resolve the tension between a monetary union, national fiscal policies and divergent competitiveness.

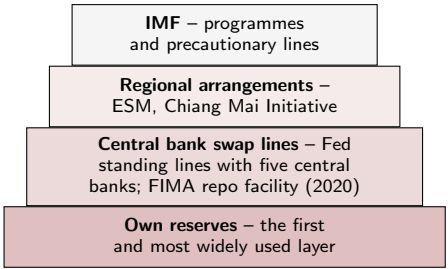
# The Policy Toolkit for Capital Flows

| Instrument | What it does | Examples and caveats |
|---|---|---|
| Exchange rate flexibility, monetary policy | First line of defence: the exchange rate absorbs shocks | Fear of floating where debt is in foreign currency |
| FX reserves and intervention | Self-insurance against stops; smooths disorderly moves | Asia's reserve build-up after 1997; costly to hold |
| Macroprudential measures (MPMs) | Limit FX lending, currency mismatches and leverage at home | Limits on FX mortgages after the CEE experience (FX lecture) |
| Capital flow management measures (CFMs) | Tax or limit inflows or outflows by residence | Chile's reserve requirement (1991–98); Brazil's tax on portfolio inflows (from 2009) |

## IMF Institutional View (2012, reviewed 2022)

CFMs should not substitute for needed macroeconomic adjustment. Since 2022, **pre-emptive** CFM/MPMs on debt inflows may be appropriate where stocks of foreign currency debt create systemic risk – targeted, temporary and transparent. The Integrated Policy Framework (Basu et al. 2020) studies how the four tools work together.

# When Flows Stop: The Global Financial Safety Net




- The net is **uneven**: advanced economies with swap lines have ample cover; many emerging markets rely on their own reserves and the IMF
- Swap lines turn a dollar sudden stop into a manageable funding problem (2008, 2020; FX lecture, slide 81)
- IMF lending comes with conditionality and stigma – one reason EMs self-insure with reserves
- In the euro area, TARGET2 and the ECB's lending play the role of the safety net inside the union

# The Czech Case: Capital Flows and the Koruna Commitment, 2013–2017

- **7 November 2013:** the ČNB commits to prevent the koruna from appreciating beyond CZK 27/EUR
- **From mid-2015:** inflows betting on the exit force heavy intervention – EUR 75 bn in total by March 2017
- Non-residents took 47% of government bond auctions in 2015–16 and 55% in the first half of 2017; short-term yields turned negative from August 2015. By 2025 non-residents held 25% of koruna government bonds (37% in 2019)
- **6 April 2017:** exit with immediate effect; the koruna appreciated gradually, without a gap

**Why the exit was orderly**

Positions were large but two-sided: exporters and investors sold the koruna as speculators bought it, so the central bank was not the only absorber (FX lecture: ČNB vs. SNB).

**Takeaway:** A credible one-way bet attracts large gross inflows into local bonds; reserves absorb them, and the exit tests liquidity and positioning.

*Source: ČNB, Inflation Report II/2017, Box 1; Inflation Report III/2017, Box 2; ČNB, Zpráva o vývoji platební bilance 2025.*

# Scaling Up: From the Euro Area to Global Dollar Risk

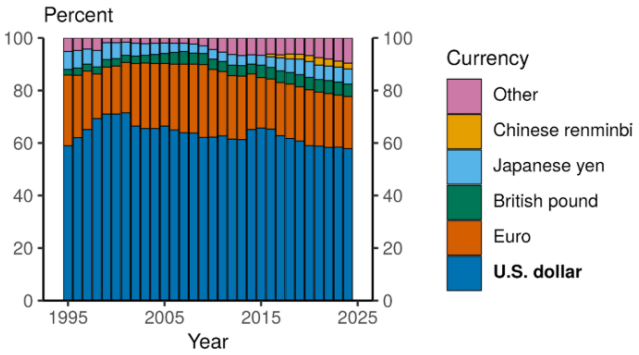
**Euro-area lesson:** sudden stops in gross flows can force painful internal adjustment when there is no exchange rate valve.

**The same risk exists globally:** the US runs persistent deficits financed by foreign gross inflows. What happens if those flows slow or reverse?

# Reserve Currencies: The Basics

- **Definition:** foreign currencies held by central banks for intervention, international payments and as a store of value
- **Dominance:** driven by the issuer's economic size, stability and network effects (inertia)
- **Key characteristics:**
  - **Liquidity:** deep, accessible financial markets
  - **Stability:** low inflation and default risk
  - **Convertibility:** unrestricted capital flows
  - **Institutional strength:** rule of law, central bank independence

# Global Foreign Exchange Reserves by Currency




**Takeaway:** The dollar's share has fallen from over 70% around 2000 to 56.7% (Q2 2026) – still more than the next three currencies combined; the renminbi remains small (2.1%).

*Source: Federal Reserve Board (2025), "The international role of the U.S. dollar – 2025 edition", FEDS Notes; IMF COFER (Q2 2026).*

# Current Status of Reserve Currencies

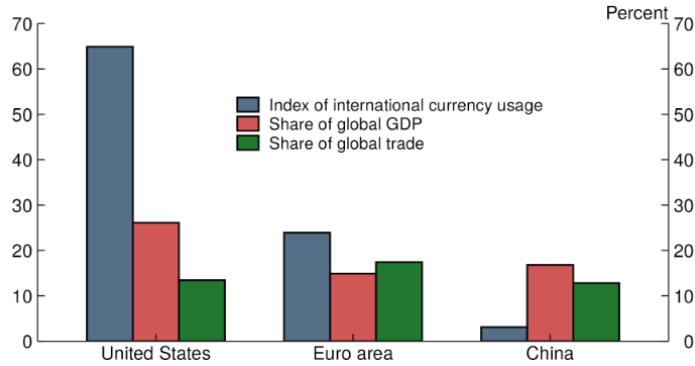
| Currency | % (Q2 2026) |
|---|---|
| US dollar (USD) | 56.70 |
| Euro (EUR) | 20.60 |
| Japanese yen (JPY) | 4.96 |
| Chinese renminbi (CNY) | 2.11 |
| Other* | 15.63 |

*GBP, CAD, AUD, CHF and others. Total reserves: USD 13.22 trillion.

- The dollar's share fell from 57.18% to 56.70% in Q2 2026, mainly because euro holdings rose
- Part of each quarter's change is valuation: when the dollar weakens, non-dollar reserves are worth more in dollars even if nobody sells dollars (IMF 2025)
- The renminbi is held back by capital controls and limited convertibility

*Source: IMF blog, "Dollar's share of reserves held steady in second quarter when adjusted for FX moves", 2025.*

# International Role vs. Size of Economy (2024)




**Takeaway:** The dollar's international use (index about 65) far exceeds the US share of world GDP (about 26%) or trade (about 13%); the euro is roughly in proportion; the renminbi far below.

*Source: Federal Reserve Board (2025), "The international role of the U.S. dollar – 2025 edition", FEDS Notes.*

# The US Dollar's Multifaceted Dominance

## Why the dollar remains the world's currency

- **Reserves:** 56.7% of allocated global reserves (IMF COFER, Q2 2026)
- **FX trading:** on one side of about 89% of all FX trades (BIS Triennial Survey 2025)
- **Trade invoicing:** about 96% of trade invoicing in the Americas (Fed 2025)
- **Payments:** around half of SWIFT payments by value
- **The core support:**
  - The deepest, most liquid **Treasury market** (safe assets)
  - **Fed swap lines** (backstop liquidity in dollar stress)
  - A strong institutional framework (**rule of law**)

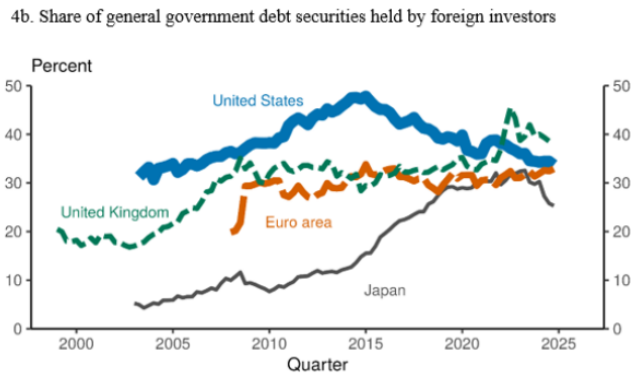
# Geopolitics and Shifting Capital Flows

The consensus around the dollar as the ultimate safe haven is being tested by the use of reserves as a sanctions weapon (Russia, 2022) and by US fiscal dynamics.

- **Declining foreign share of US debt:** foreign investors hold about a third of US government debt securities, down from almost half in 2014 (next slide)
- **The rise of gold:** at market prices, foreign central banks' gold overtook their US Treasury holdings in 2025 – for the first time since 1996; gold was about 20% of global official reserves at end-2024, ahead of the euro (16%) (ECB 2025)
- **Fiscal dominance risks:** US net interest outlays ($881 bn in fiscal 2024, CBO) now exceed defence spending; markets increasingly ask whether Fed independence will hold
- **April 2025:** the dollar fell while US yields rose – unusual for a safe haven; FX lecture, slide 34, shows the hedging channel

**Implication:** global capital flows are fragmenting into blocs; deficit countries such as the US can no longer take foreign financing for granted.

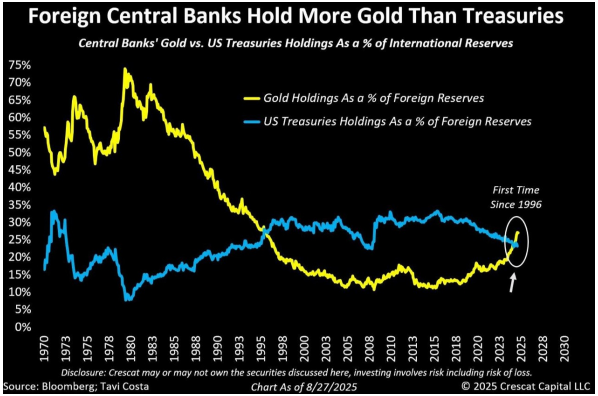
# Government Debt Held by Foreign Investors




**Takeaway:** Foreigners hold about a third of US government debt, down from almost half in 2014 – similar to the euro area and the UK; the US now relies more on domestic buyers.

*Source: Federal Reserve Board (2025), "The international role of the U.S. dollar – 2025 edition", FEDS Notes.*

# The Rise of Gold




**Takeaway:** At market prices, gold overtook Treasuries in foreign central banks' reserves in 2025 for the first time since 1996 – driven by purchases of over 1,000 tonnes a year in 2022–24 and by the gold price.

*Source: Crescat Capital, data: Bloomberg; ECB (2025), The international role of the euro, June.*

# Conclusions (I): Vulnerabilities vs. Triggers

- **Accounting realities (the vulnerability):** current account deficits reflect saving–investment imbalances, $(S - I) + (T - G)$. They are a *medium- to long-term vulnerability gauge*, not an immediate crisis trigger
- **Look beyond the net:** financial crises are rarely caused by net trade imbalances. They are fuelled by the build-up of **gross capital flows** (bank lending, wholesale funding, portfolio flows), enabled by elastic credit creation
- **Sudden stops (the trigger):** crises are ignited by abrupt reversals in volatile gross flows, which create immediate liquidity and solvency problems and expose the underlying vulnerabilities

# Conclusions (II): Policy, Distortions and Tail Risks

- **Policy:** monitor gross positions, currency mismatches and non-resident holdings; combine exchange rate flexibility, reserves, macroprudential measures and – where justified – CFMs; the global safety net is uneven
- **Asset market distortions:** persistent imbalances sustain asset booms and non-productive investment; large equity portfolio flows move valuations far from domestic fundamentals
- **Structural fragility and leverage:** dependence on foreign financing gives reserve holders leverage – for example by diversifying away from Treasuries
- **The geopolitical limit:** the dollar's exorbitant privilege is being tested; large fiscal deficits in a fragmenting world raise tail risks if foreign gross inflows stop

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